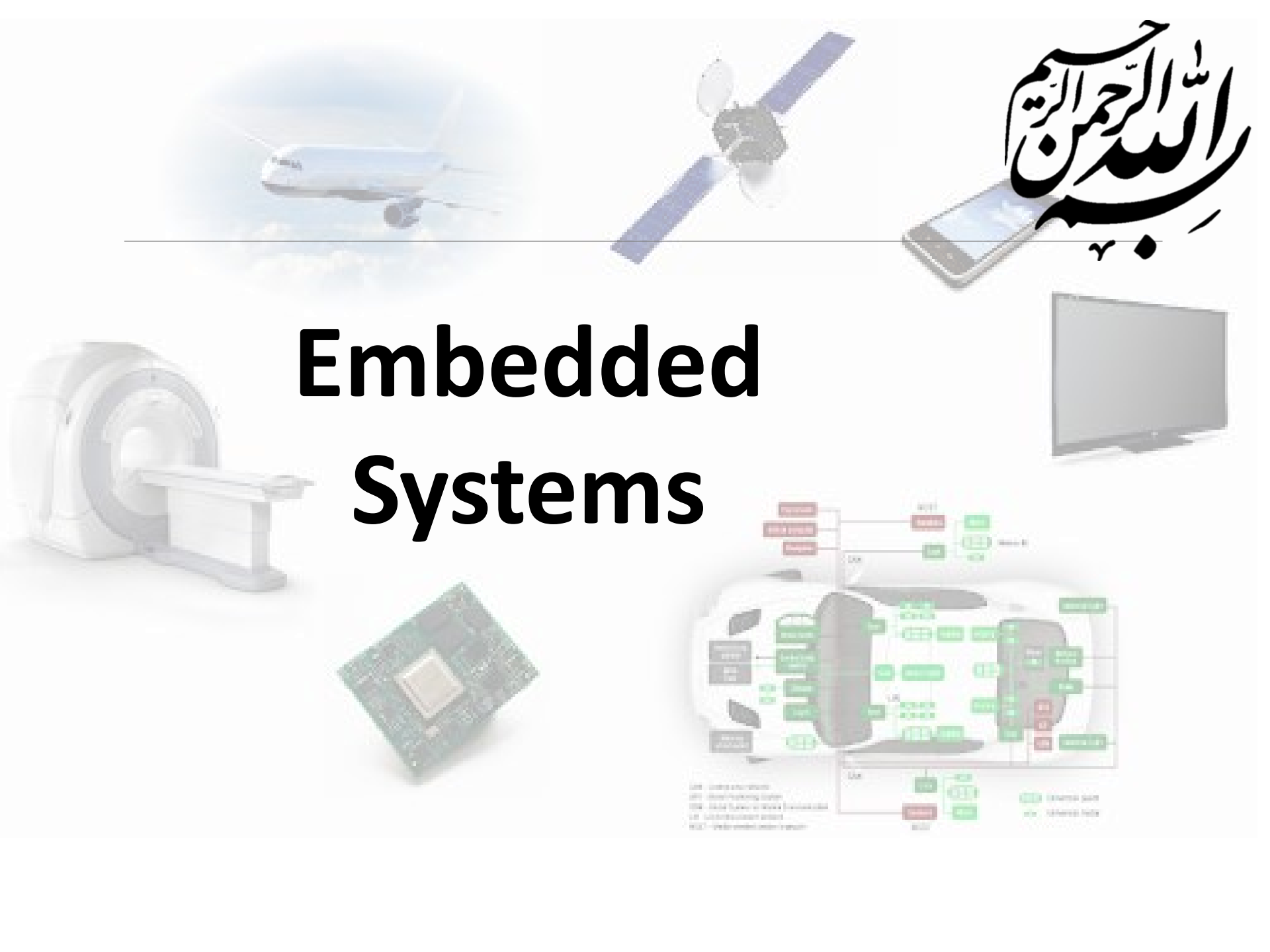


السلامة

Embedded Systems



Lecture1:

Introduction

References

Lee EA, Seshia SA. Introduction to embedded systems: A cyber-physical systems approach. MIT press; 2017 Jan 6.

Marwedel P. Embedded systems foundations of cyber-physical systems, and the internet of things. Embedded System Design; Springer Nature: Cham, Switzerland. 2018.

M. Wolf, Computers as Components Principles of Embedded Computing System Design, 2023.

C. Ünsalan, H. D. Gürhan and M. E. Yücel, Embedded System Design with ARM Cortex-M Microcontrollers Applications with C, C++ and MicroPython, 2022.

R. Banerjee, Hands-on TinyML Harness the power of Machine Learning on the edge devices, 2023.

Evaluation

- **Assignments:** 4 Point
- **Midterm:** 8 Point
- **Final exam:** 8 Point
- **Final project:** 1 Point

Outline

- **Basic definitions**
- **Embedded system components**
- **Microcontrollers**
- **Design process**
- **A review of artificial neural network and deep learning**
- **A review of Python**
- **Introduction to TinyML**
- **Model Optimization algorithms**
- **Simple projects**

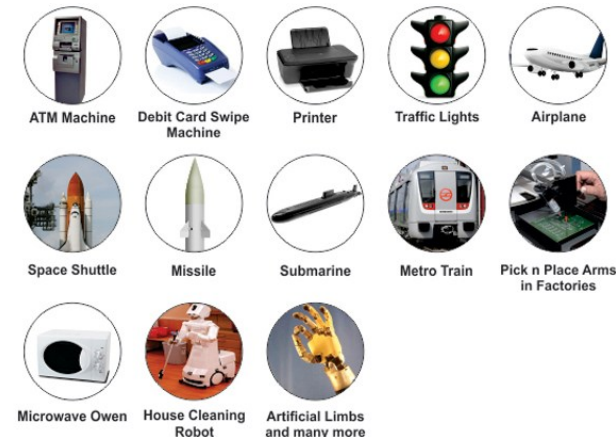
Basic definitions

- **Computing systems**
- **What's an Embedded system?**

Basic definitions

➤ Computing systems

- Large mainframe computers (before 1950)
- Personal computers (PCs) (1970)
- Mark Weiser's prediction: "According to Weiser, the computer as a dedicated device should disappear, while at the same time making its information processing capabilities available throughout our surroundings."
- "ubiquitous computing": the computer for the 21st century-computing **anytime, anywhere invisible** computer



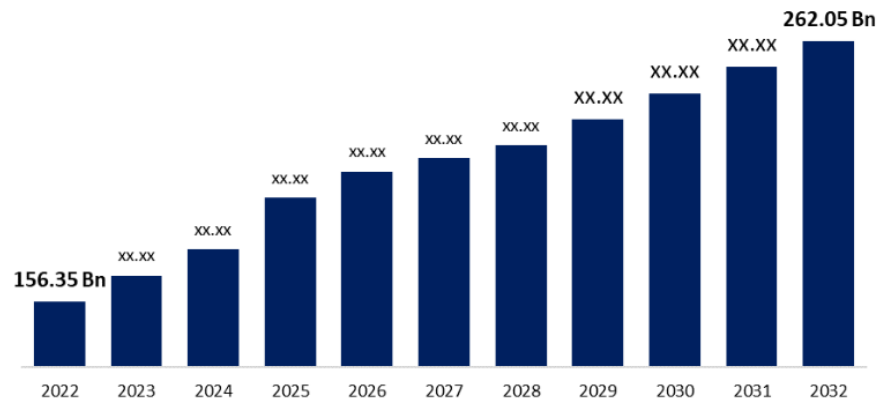
Basic definitions

➤ Computing systems

Two types of computing systems:

- Non-Embedded: PCs, Servers, and notebooks→ produced millions/year
- Embedded: the future of computing→ billions/year

Global Embedded System Market Size



Global Embedded System Market worth \$262.05 billion by 2032

Basic definitions

➤ Embedded systems

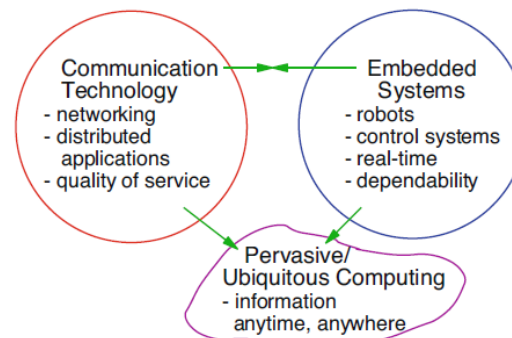
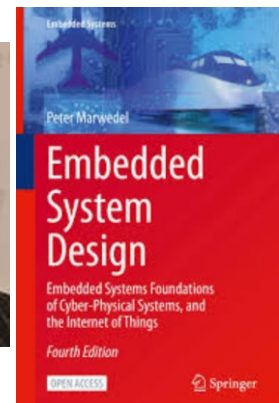
- Devices other than desktop PCs, servers, and notebooks
 1. Electricity running through
 2. Perform something intelligent

■ Embedded systems are information processing systems embedded into enclosing products, (Marwedel).

Ex: cars. Trains, planes, and telecommunication

■ Common characteristics

1. Real-time constraints
2. Dependability requirements
3. Efficiency requirements



Basic definitions

➤ Embedded systems

- We expect our devices to be **capable**. They must be feature-rich, providing many options on what they can do and how to do them.
- They must be **very efficient**.
- Battery-powered devices must go a long time between charges.
- They must be **physically small**, even though they are capable.
- They must be **inexpensive**.
- They must be **reliable**. They must work as expected without failure for long periods. If they do break, they must do so safely, and they should adjust themselves as they operate to maintain their accuracy and effectiveness.
- They must be **Real-time and concurrent execution**. Embedded computers and software need to operate in real time. The world doesn't stop for the computer to catch up.

Basic definitions

➤ Cyber-physical systems (cps):

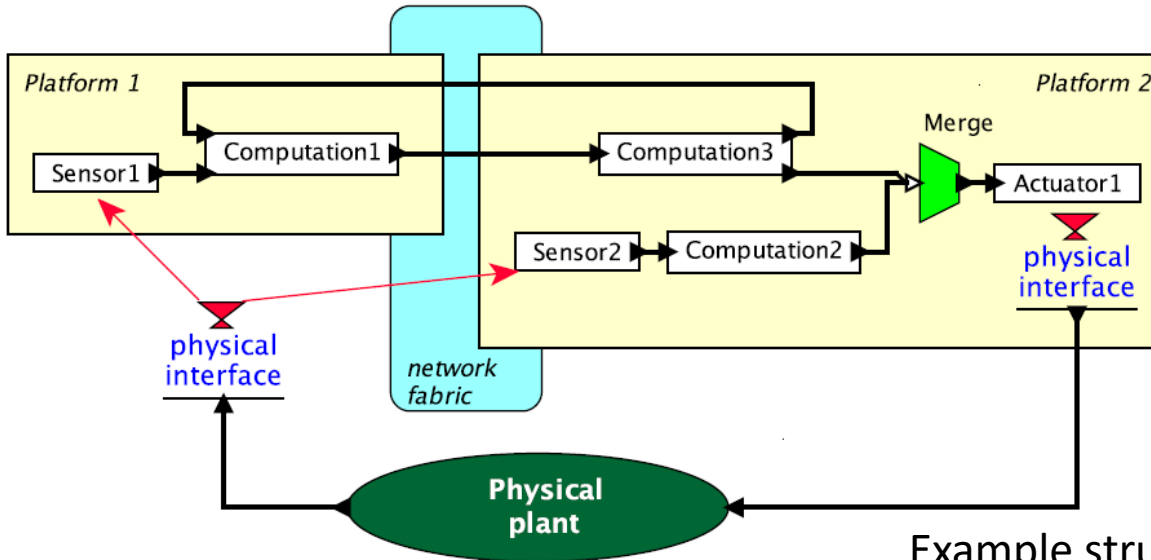
- Cyber-physical systems (cps) are integrations of computation and physical processes (E. A. Lee and S. A. Seshia).

CPs = ES + (dynamic) physical environment

Cyber-physical system (CPS)

Embedded system

Physical environment

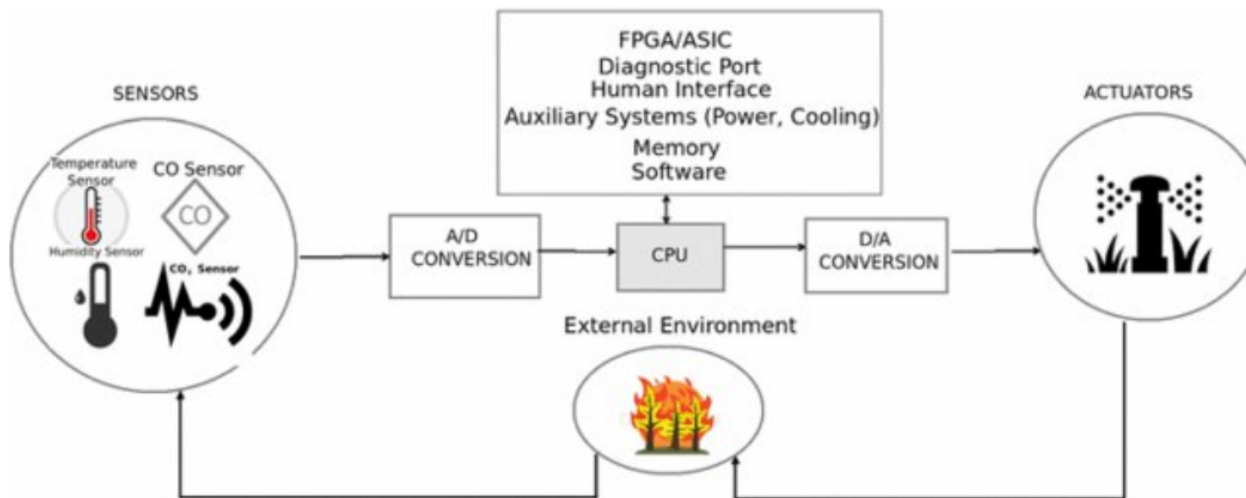


+ artificial intelligence → changed our life
The future of Computing Systems

Example structure of a cyber-physical system.

Basic definitions

➤ Embedded systems example:



Components of embedded systems

- Analog components
 - Sensors, actuators,...
- Digital components
 - Processors
 - Memories
 - Buses
 - Application Specific Integrated Circuits (ASIC)
- Converters- A/D, D/A,...
- Software
 - Application programs
 - Exception Handlers

Components-processors

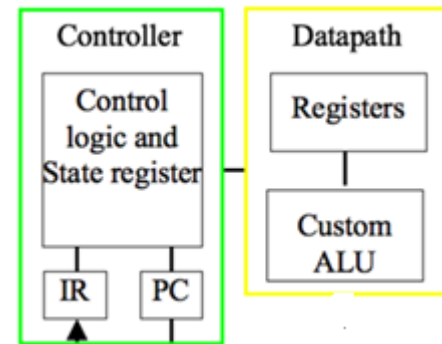
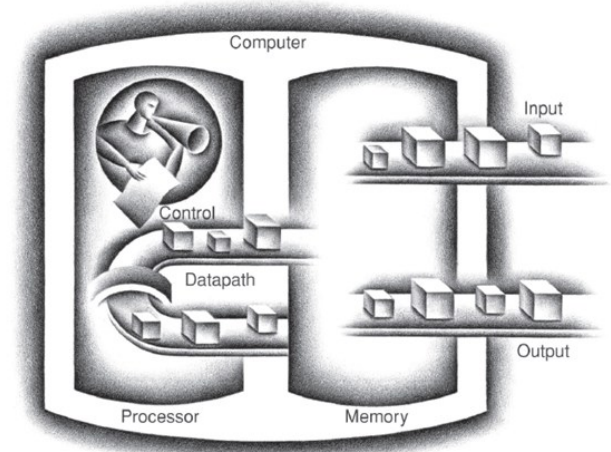
➤ Processors

What is a processor?

- Artifact that computes (runs algorithms)
- Controller and data-path

GP vs SP processors

- General purpose processors
- Specific (single) purpose processors



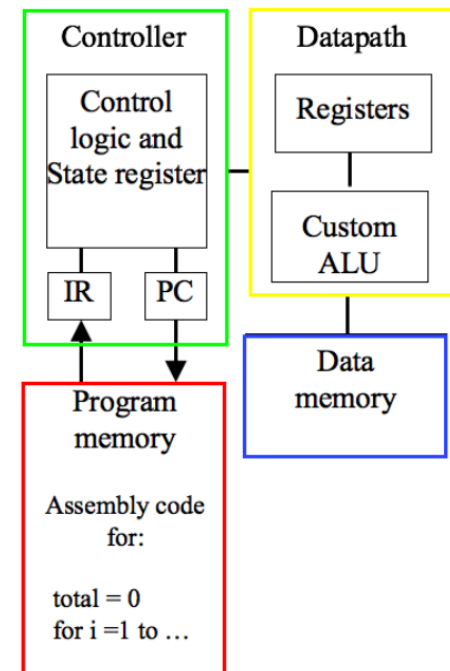
Components-processors

- General-purpose (GP) processors:
 - Variety of computation tasks
 - Functional flexibility and low cost at high volume (maybe)
 - Slow and power hungry
- Single-purpose (sp) processors (or ASIC)
 - One particular computation task
 - Fast and power efficient
 - Functional inflexibility and high cost at low volumes (maybe)

Components-processors

➤ General purpose processors:

- Programmable device used in variety of applications
also known as “microprocessor”
- Features
 1. Program memory
 2. General datapath with large register file and general Alu
- User benefits
 1. Low time-to-market
 2. High flexibility develop
- Examples
Pentium, Power PC



Components-processors

➤ Application Specific IS Processors (ASIPs)

- Programmable processor optimized for a particular class of applications having common characteristics
- Compromise between general-purpose and ASIC (custom hardware)

■ Features

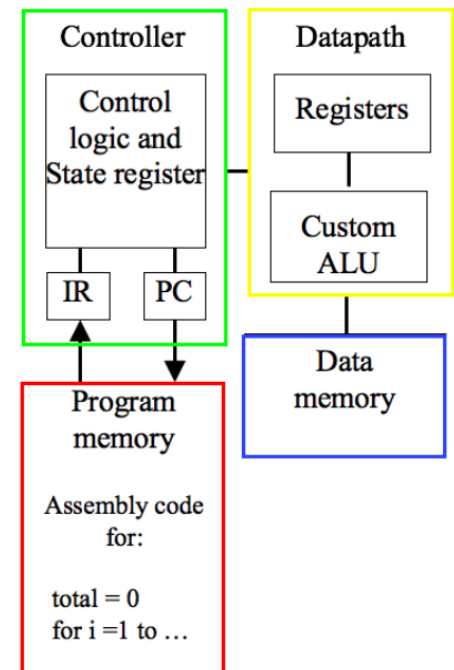
1. Program memory
2. Optimized datapath
3. Special functional units

■ Benefits

Some flexibility, good performance, size and power

■ Examples

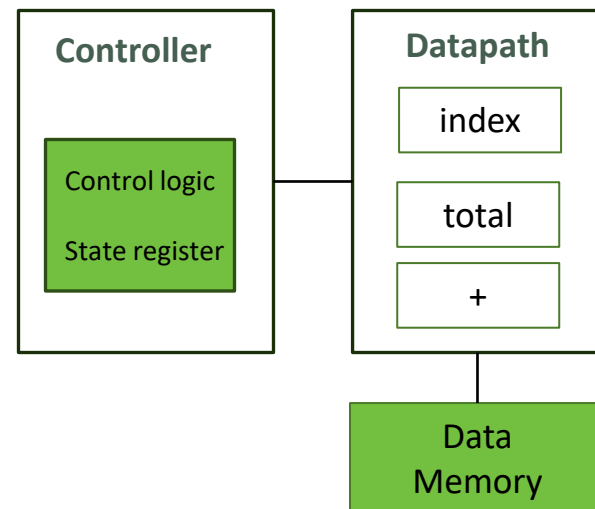
DSPs, video signal processors, Network processors,...



Components-processors

Application-Specific ICs (ASICs):

- Digital circuit designed to execute exactly one program
Coproprocessor, hardware accelerator
- Features
 - Contains only the components needed to execute a single program
 - No program memory
- Benefits
 - Fast
 - Low power
 - Small size
- Examples
Digital voice recorders, digital currency miners,...



Components-processors

➤ GP vs. SP processors

GP:

- Programmable controller
 - Control logic is stored in memory
 - Fetch/decode overhead
- Highly general data-path
 - Typical bit-width (8, 16, 32, 64)
 - Complete set of arithmetic/logic units
 - Large set of registers

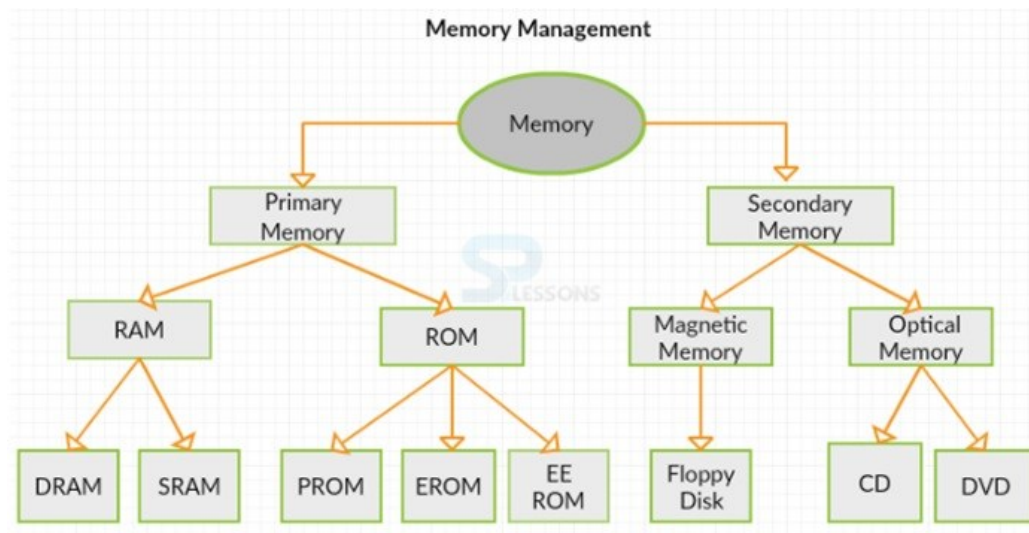
ASIC:

- Hardwired controller
 - No need for program memory and cache
 - No fetch/decode overhead
- Highly tuned data-path
 - Custom bit-width
 - Custom arithmetic/logic units
 - Custom set of registers

Components-storage

➤ Storage:

- What is a memory?
 - Artifact that stores bits
 - Storage fabric and access logic



Components-storage

➤ Storage:

- Write-ability
Manner and speed a memory can be written
- Storage-permanence
Ability of memory to hold stored bits after they are written
- Many different types of memories
Flash, SRAM, DRAM, etc.

Components-storage

➤ Write-ability

Ranges of write ability

■ High end

- Processor writes to memory simply and quickly
- E.g., RAM

■ Middle range

- Processor writes to memory, but slower
- E.g., FLASH, EEPROM

■ Lower range

- Special equipment, “programmer” must be used to write to memory
- E.g., EPROM, ROM

■ Low end

- Bits stored only during fabrication
- E.g., mask-programmed ROM

Components-BUS

➤ Interface: BUS and communication:

- Information must be communicated before it can be processed in an embedded system.
- Information can be communicated through various channels.
- Channels are abstract entities characterized by the essential properties of communication (maximum information transfer capacity, noise,...).

Components-BUS

➤ **Interface: BUS and communication:**

- The physical entities enabling communication are called communication media
- Important media classes include:
 - Wireless media (radio frequency media, infrared)
 - Optical media (fibers)
 - wires

Components-BUS

Interface: BUS and communication: requirements

- There is a huge variety of communication requirements between the various classes of embedded systems.
- Some common requirements can be identified

Components-BUS

➤ **Interface: BUS and communication:** requirements

- **Real-time behavior**

Several low-cost solutions such as standard Ethernet fail to meet this requirement.

- **Appropriate bandwidth and communication delay:**

Bandwidth requirements of embedded systems may vary. It is important to provide sufficient bandwidth without making the communication system too expensive.

- **Efficiency**

Connecting different hardware components can be expensive.

- **Security/privacy**

Ensuring security/privacy of confidential information (confidentiality) may require the use of encryption.

Components-BUS

➤ **Interface: BUS and communication: requirements**

- **Safety/robustness**

For safety-critical systems, the required level of safety must be achieved.

This includes robustness: for example: cyber-physical systems may be used at extreme temperatures. Close to major sources of electromagnetic radiation. Etc. car, engines,...

- **Fault tolerance**

Despite all the efforts for robustness, faults may occur cyberphysical systems should be operational even after faults if at all feasible.

Restarts, like the ones found in pcs, cannot be accepted retries may be required after attempts to communicate failed.

A conflict exists within first requirement: if we allow retries then it is difficult to meet real-time requirements.

- **Maintainability, diagnosability:**

It should be possible to repair embedded systems within reasonable time frames.

Components-BUS

➤ Interface: BUS and communication:

NOTE:

- Embedded systems use interfaces for communication
- Interfaces are implemented using Bus structures.
- Many types of interfaces are used in ES (vs PCs), I^2C , SPI,...

I2C stands for Inter-Integrated Circuit
SPI stands for Serial Peripheral Interface

Components-BUS

➤ Interface: BUS and communication:

What is a bus?

- An artifact that transfers bits (Implementation of communication media)
- Media (physical channel): wires, air, or fiber
- Interface logic: for controlling media and communication

Components-BUS

➤ Interface: BUS and communication:

- Connectivity scheme
 - Serial communication
 - Parallel communication
 - Wireless communication
- Protocol
 - Ports
 - Timing diagrams
 - Read and write cycles
- And:
 - Arbitration scheme
 - Error detection/correction
 - DMA
 - ...

→ Standard of communication

Components-BUS

➤ Interface: BUS and communication:

■ Serial communication

- A single wire used for data transfer
- One or more additional wires used for control
 - Some protocols may not use additional control wires
- Higher throughput for long distance communication
 - Often across processing node
- Lower cost in terms of wires (cable)
- E.g., USB, Ethernet, RS232, I^2C , etc.

Components-BUS

➤ Interface: BUS and communication:

■ Parallel communication

- Multiple buses (wires) used for data transfer
- One or more additional used for control
- Higher throughput in short distance communication

Data misalignment problem

Often used within a processing node

- Higher cost in terms of wires (cable)
- E.g., ISA, PCI, etc.

Industry Standard Architecture (ISA) is the **16-bit** internal bus of IBM

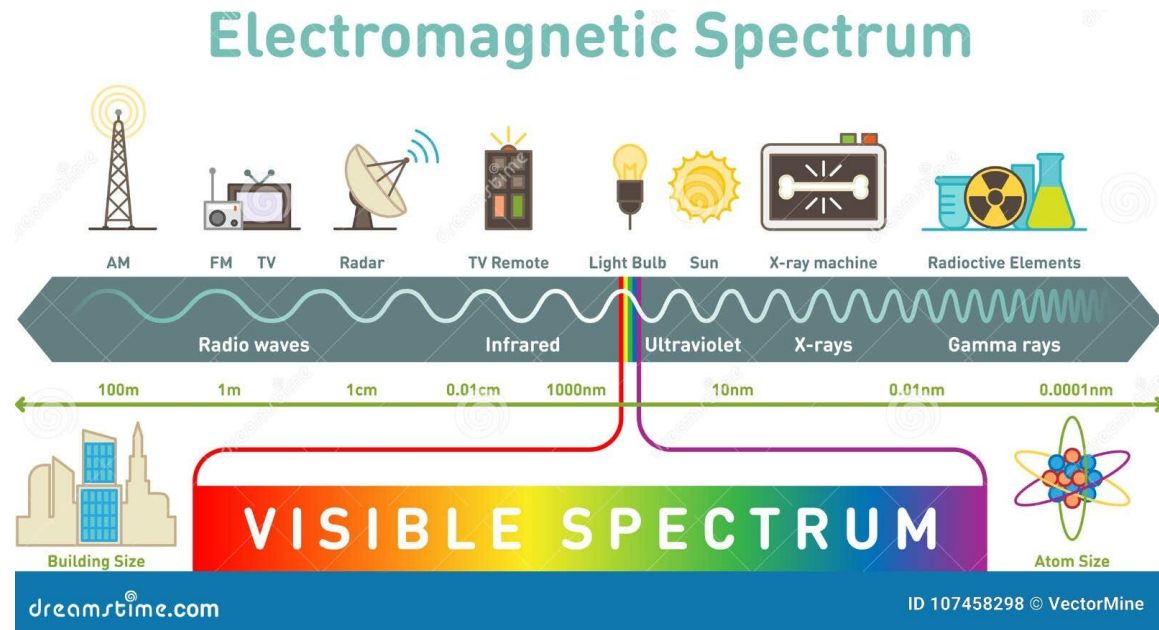
Peripheral Component Interconnect (PCI) is a **local computer bus**

Components-BUS

➤ Interface: BUS and communication:

■ Wireless communication (Electromagnetic):

- Infrared radiation (IR)
- Radio frequency (RF)



Components-BUS

➤ Interface: BUS and communication:

■ Wireless communication Infrared (IR):

- Electromagnetic wave frequencies just below visible light spectrum
- Diode emits infrared light to generate signal
- Infrared transistor detects signal, conducts when exposed to infrared light
- Cheap to build
- Need line of sight, limited range

■ Wireless communication: Radio Frequency (RF):

- Electromagnetic wave frequencies in radio spectrum
- Analog circuitry and antenna needed on both sides of transmission
- Line of sight not needed, transmitter power determines range.

